



Effects of Chinese College Students' Academic Integrity Perceptions on the Intention To Disclose AI-Generated Outputs: A Moderated Mediation Model

Xiaoqin Zhang¹ · Jinbao Zhang¹ · Mohamed Oubibi¹ 

Accepted: 24 March 2025 / Published online: 7 April 2025
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract

Academic integrity is a critical ethical issue in the use of generative artificial intelligence (AI) for writing assistance. Disclosure of AI-generated content is essential to maintain the authenticity and reliability of information. This study employs a moderated mediation model to examine the mediating role of AI ethics attention in the relationship between academic integrity perceptions and the intention to disclose AI-generated outputs, as well as the moderating effect of AI usage frequency. A survey was conducted among 519 valid respondents across 28 provinces, municipalities, and special administrative regions in China. The findings indicate that college students' perceptions toward academic integrity do not directly influence their intention to disclose AI-generated content. Instead, AI ethics attention mediates the relationship between academic integrity perceptions and disclosure intentions. Furthermore, AI usage frequency negatively moderates both the relationship between AI ethics attention and disclosure intentions, and the mediating role of AI ethics attention in the link between academic integrity perceptions and disclosure intentions.

Keywords College students · Academic integrity · Generative artificial intelligence · AI ethics · Disclosure of AI-generated outputs

Introduction

The rapid development of generative artificial intelligence (AI) is profoundly transforming the way college students acquire knowledge (Bahroun et al., 2023; Dai et al., 2023; Oubibi et al., 2025). As a transformative educational resource, generative AI not only enriches the interaction between teachers and students but also brings a series of positive impacts to higher education (Chan & Tsi, 2024; Shahzad et al., 2024). Research shows that generative AI can enhance learning efficiency, enjoyment, satisfaction, and curiosity among college students,

✉ Mohamed Oubibi
oubibi@bnu.edu.cn

¹ Faculty of Education, Smart Learning Institute of Beijing Normal University, Beijing Normal University, Beijing 100082, China

thus stimulating their task motivation (Hmoud et al., 2024), improving critical thinking and problem-solving abilities (Kasneji et al., 2023), and through effective interactions with generative AI tools, students' academic performance is significantly improved (Liang et al., 2023; Oubibi et al., 2024). However, as the application of generative AI becomes more widespread, ethical and academic integrity issues are becoming increasingly prominent (Eke, 2023). For example, college students may misuse these tools to complete assignments, leading to issues such as assignment cheating, academic plagiarism, and other ethical and academic integrity concerns (Dawson, 2020; Mulenga & Shilongo, 2024; Perkins et al., 2020). Eaton (2023) point out that we are in what could be called a "post-plagiarism" era, where artificial intelligence and neurotechnology have become deeply embedded in our daily lives (Eaton, 2023). AI can now generate content that rivals or even surpasses human-created content in quality, which blurs the lines between originality and plagiarism and makes it difficult to detect. More importantly, generative AI has the potential to produce misinformation and disinformation (Lo, 2023; Tlili et al., 2023), posing significant challenges to ensuring the truthfulness and reliability of information. The transparency and disclosure of AI-generated content are crucial for maintaining the integrity of information and communication (Wang et al., 2023). Therefore, to preserve the authenticity and reliability of information, the use of AI in content generation must be disclosed (El Ali et al., 2024). However, studies show that college students struggle to differentiate between "proper" and "improper" uses of generative AI, leading to ethical ambiguity when determining what constitutes personal original work versus content assisted by AI (Duah & McGivern, 2024). Nonetheless, individuals still bear ultimate responsibility for their work, including ensuring the accuracy and originality of academic content and clearly defining the contributions of all parties involved. Students and researchers need to clearly mark which parts of their work were generated by AI and which are their own original contributions (Eaton, 2023). In this regard, OpenAI specifically emphasized "providing transparency for AI-generated content" in its statement for the 2024 global elections (Openai,). To maintain the authenticity and reliability of information and increase transparency in AI-assisted writing, various academic institutions and publications both domestically and internationally have established clear guidelines for disclosing AI-generated outputs. For example, the National Natural Science Foundation of China's "Handbook of Scientific Research Integrity" requires disclosure of the manner and main details of using generative AI in appropriate sections such as research methods or appendices. For content generated by generative AI, explicit labeling and description of the generation process is required. The European Union's Artificial Intelligence Act specifies transparency requirements for generative AI, such as mandating that deployers of AI systems that generate or manipulate deepfake images, audio, or video content must disclose that these contents are artificially generated or manipulated (Act,). Among the 100 largest academic publishers and journals internationally, 42.8% of publishers and 44.3% of journals have established specific disclosure standards for the use of generative AI (Ganjavi et al., 2023). Bozkurt (2024) advocates for transparent reporting methods in AI assisted academic writing to maintain academic integrity, and proposes a framework called Academic Integrity and Transparency in AI assisted Research and Specification (aiTARAS) to acknowledge and disclose the use of generative artificial intelligence in academic writing to ensure compliance with academic integrity and ethical standards (Bozkurt, 2024).

Based on the above context, understanding how college students balance technology application with academic integrity becomes particularly important. Specifically, with the

growing popularity of generative AI tools in assisting academic writing, the appropriate disclosure of AI-generated content has become a focal point of concern in academia. This study focuses on this critical issue, aiming to explore whether college students' intentions to disclose AI-generated outputs are influenced by their personal concepts of academic integrity, and to reveal the complex relationship between students' disclosure intentions and their academic integrity perceptions. To this end, this study primarily addresses the following research questions:

RQ1. Do college students' perceptions of academic integrity directly predict their intention to disclose AI-generated outputs?

RQ2. How do college students' perceptions of academic integrity influence their intention to disclose AI-generated outputs through the mechanism of AI ethics attention?

RQ3. How does the frequency of AI usage affect the relationship between AI ethics attention and the intention to disclose AI-generated outputs?

Theoretical Background and Hypotheses Development

Academic Integrity Perceptions and Intention To Disclose AI-Generated Outputs

Academic integrity does not have a single, universally accepted definition within academia. The International Center for Academic Integrity (Macfarlane et al., 2014) defines academic integrity as a commitment to six core values: honesty, trust, fairness, respect, responsibility, and courage. Rodrigues et al. (2024) view academic integrity as a fundamental principle based on honesty, responsibility, transparency, and trust (Rodrigues et al., 2024). For the purposes of this study, academic integrity is defined as adherence to a set of ethical standards and behavioral guidelines in academic activities. Specifically, behavioral guidelines include practices such as submitting original work, properly citing sources, avoiding plagiarism, disclosing any assistance received, and ensuring impartiality in academic assessments and collaborations. It requires students and scholars to demonstrate honesty, fairness, respect for intellectual property rights, and accountability in their academic work. The concept of academic integrity perception refers to an individual's recognition and attitude toward the importance of academic integrity, reflecting one's academic value orientation and behavioral guidance. The disclosure of AI-generated outputs refers to the process where authors explicitly indicate which parts of their academic works or other content creations were generated or assisted by artificial intelligence systems. The intention to disclose AI-generated outputs is the psychological inclination or plan that an author has to clearly identify AI-generated content in their final work when using AI-assisted creation.

Social Cognitive Theory posits that an individual's behavioral tendencies are influenced by their beliefs and values (Bandura, 1988). Therefore, college students who hold strong perceptions of academic integrity are more likely to believe that appropriately disclosing AI-generated content is their responsibility. This sense of responsibility would motivate them to be more inclined to disclose such information when submitting works that contain

AI-generated content. Techno ethics emphasizes the importance of ethical considerations during the use of technology (Stahl & Eke, 2024), and students with strong academic integrity perceptions are likely to also highly value transparency and authenticity when using AI tools, making them more prone to voluntarily disclose which parts of their work were produced by artificial intelligence. Thus, this study proposes Hypothesis H1: College students' perceptions of academic integrity have a positive influence on the intention to disclose AI-generated outputs.

The Mediating Role of AI Ethics Attention

To provide more direct implications for intervention practices, it is necessary to further explore the mediating process through which college students' perceptions of academic integrity influence their intention to disclose AI-generated outputs. Based on existing literature and relevant theories, this study posits that AI ethics attention may serve as an important mediating variable between academic integrity perceptions and the intention to disclose AI-generated outputs. The following paragraph provides a detailed discussion of these studies and their relevance to Hypothesis 2.

AI ethics attention refers to the degree of concern individuals have regarding ethical issues when using AI tools. In a higher education setting, students not only need to acquire knowledge and skills specific to their field but also should be mindful of potential ethical issues during their use of technology and maintain good moral judgment. According to the perspective of technological ethics, individuals with a strong sense of academic integrity tend to exhibit higher levels of AI ethics attention when using AI tools, as they recognize the importance of adhering to ethical standards in academic communication. Furthermore, when they demonstrate high levels of AI ethics attention, they are more likely to have a strong intention to disclose after using AI tools, because doing so aligns with their emphasis on academic integrity. Cognitive dissonance theory posits that when an individual detects a contradiction between two or more cognitions they hold, they experience a psychological discomfort or tension known as cognitive dissonance (Festinger, 1957). To reduce this dissonance, individuals typically take steps to adjust their attitudes or behaviors to achieve consistency and restore psychological equilibrium. According to cognitive dissonance theory, those who have a strong commitment to academic integrity may increase their AI ethics attention to maintain consistency in their behavior, which in turn can promote the intention to disclose AI-generated outputs. A strong sense of academic integrity prompts individuals to consider the ethical implications of using AI tools, thus increasing their AI ethics attention. The key contradiction here is between the value of academic integrity and the act of using AI-generated content without acknowledging its contribution. High AI ethics attention makes individuals more likely to realize that failing to disclose AI-generated content would conflict with their values of academic integrity, leading to cognitive dissonance. To reduce this dissonance, individuals are inclined to disclose their use of AI, thereby upholding their academic integrity and reinforcing their intention to disclose AI-generated outputs.

Based on the above argument, this study proposes Hypothesis H2: AI ethics attention mediates the relationship between academic integrity values and the intention to disclose AI-generated outputs.

The Moderating Role of AI Usage Frequency

AI usage frequency indicates how often individuals use AI tools. This measure of dependence on AI significantly affects their behavior and attitudes, moderating the link between AI ethics attention and disclosure intentions. Habituation theory supports this, stating that repeated exposure to a stimulus weakens response over time (Colwill et al., 2023; Waters & McDonald, 1976). Under certain conditions, this can lead to over-habituation, causing a persistent decrease in response or failure to recover (Waters & McDonald, 1976).

For college students, frequent AI use may habituate them to AI ethics issues and outputs, reducing their initial high ethical attention and disclosure intentions. Initially, unfamiliarity with AI may prompt high ethical attention and transparency. However, regular use can lead to habituation, decreasing attention to AI ethics and disclosure motivation.

Cognitive adaptation theory suggests that long-term AI use adjusts individuals' cognitive systems, diminishing awareness of ethical issues. This adaptation, or "switching costs" can reduce responsiveness to new stimuli (Dang et al., 2014). Frequent AI use may focus students on tool functionality and operational efficiency, not ethics, weakening the link between AI ethics attention and disclosure.

In summary, increased AI usage may reduce students' ethical attention and disclosure intentions through habituation or adaptation. This suggests AI usage frequency moderates the impact of academic integrity perceptions on disclosure intentions. Hence, the study proposes Hypothesis **H3**: AI usage frequency negatively moderates the relationship between AI ethics attention and the intention to disclose AI-generated outputs. That is, an increase in AI usage frequency weakens the positive effect of AI ethics attention on the intention to disclose AI-generated outputs.

In summary, this study constructs a theoretical hypothesis model illustrating how college students' perceptions of academic integrity influence their intention to disclose AI-generated outputs, as shown in Fig. 1.

Methods

Research Tool

A self-designed questionnaire was used in order to be applicable to Chinese students. For the full questionnaire, please refer to the Appendix 1. The questionnaire is divided into five

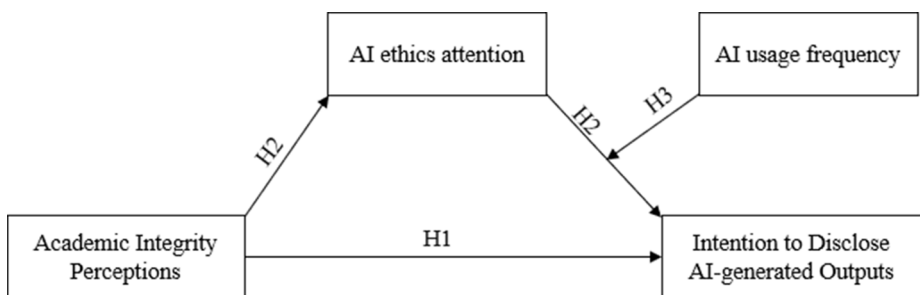


Fig. 1 Hypothetical model of this study

main sections. The first section gathers demographic data like gender, year of study, education level and discipline. The second, using a 5-point Likert scale, assesses perceptions of academic integrity. The third measures the frequency of using generative AI tools. The fourth evaluates awareness of ethical issues related to generative AI. The fifth examines the intention to disclose AI-generated content, with scenarios and opinions on acceptability.

To ensure that the questionnaire accurately reflects the research objectives and has content validity, we invited two experts to review the questionnaire. Both experts are faculty members at a double-first-class university in China: one is a professor specializing in higher education with a focus on artificial intelligence in education, and the other is an associate professor in educational technology, specializing in questionnaire design for academic studies. After two rounds of revisions and feedback, an initial version of the questionnaire was formed. In early March 2024, a pilot test was conducted on 60 university students. Through exploratory factor analysis, items with low factor loadings or those weakly related to the factors (such as standardized loading coefficients with absolute values less than 0.6) were removed, resulting in the final questionnaire. The final version of the academic integrity perceptions questionnaire consisted of four items that loaded onto a single common factor, covering aspects such as adherence to academic standards, respect for originality, compliance with citation norms, and recognition of academic responsibility. The internal consistency coefficient of this questionnaire was 0.802, indicating high reliability and good inter-item consistency. Furthermore, confirmatory factor analysis (CFA) was performed on the academic integrity perceptions scale to examine the construct validity of the questionnaire. The results showed that all item loadings had absolute values greater than 0.6 and were statistically significant, suggesting good measurement relationships between the items and the factor. The average variance extracted (AVE) for the academic integrity perceptions factor was 0.513, which exceeds the threshold of 0.5, and the composite reliability (CR) was 0.807, exceeding the criterion of 0.7, indicating good convergent validity. All model fit indices were within acceptable ranges ($\chi^2/df=1.919<3$, $GFI=0.996>0.9$, $RMSEA=0.042<0.1$, $RMR=0.005<0.05$, $CFI=0.997>0.9$, $NFI=0.994>0.9$, $NNFI=0.991>0.9$), suggesting that the scale has good structural validity.

Data Collection

Quantitative Data Collection

The target population for this study consists of college students, including full-time undergraduate and postgraduate students. The main method of data collection was through an online survey conducted on Wenjuanxing, a popular online survey platform in China. Researchers and survey administrators distributed the survey link in various WeChat groups. The online survey was conducted from March 24 to March 31, 2024, and a total of 519 electronic questionnaires were received from 28 provinces, municipalities, and special administrative regions in China. After screening the questionnaires, all 519 questionnaires were found to be valid, resulting in a 100% validity rate. The statistical information for the study sample is presented in Table 1.

In the study sample, in terms of gender, female participants had the highest representation, followed by males, with a smaller proportion of participants identifying as other genders or choosing not to disclose their gender. Specifically, males accounted for 31.21%,

females accounted for 62.43%, other genders accounted for 1.16%, and those who chose not to disclose their gender accounted for 5.20%. In terms of educational level, the majority of participants were undergraduates, accounting for 81.1%, while postgraduates accounted for 18.9%. Regarding year of study, first-year students were the most represented, with 30.44%, followed by postgraduate students at 18.88%, second-year students at 21.77%, third-year students at 12.72%, and fourth-year students at 16.18%. In terms of academic discipline, social sciences had the highest representation at 31.79%, followed by engineering and technology at 30.06%, humanities and arts at 17.53%, and natural sciences at 8.29%. In terms of school distribution, more than half (51.8%) of the survey participants came from non-double-first-class universities, while students from double-first-class universities accounted for nearly half (48.2%).

Qualitative Data Collection

To gain deeper insights into the quantitative findings, we conducted an additional focus group interview with five college students from a Chinese university, selected through convenience sampling across diverse majors (e.g., information management, logistics management, data science, and e-commerce). The 50-minute face-to-face session utilized semi-structured questions focused on students' ethical dilemmas in using AI tools academi-

Table 1 Statistical information of the study sample

Category	item	Frequency	Percentage	Valid Percentage	Cumulative Percentage
Gender	Males	162	31.21	31.21	31.21
	Females	324	62.43	62.43	93.64
	Other genders	6	1.16	1.16	94.80
	Chose not to disclose	27	5.2	5.2	100
Educational level	Undergraduates	421	81.1	81.1	81.1
	Postgraduates	98	18.9	18.9	100
Year of study	First-year	158	30.44	30.44	30.44
	Second-year	113	21.77	21.77	52.22
	Third-year	66	12.72	12.72	64.93
	Fourth-year	84	16.18	16.18	81.12
	Postgraduate	98	18.88	18.88	100
Academic discipline	Natural Sciences	43	8.29	8.29	8.29
	Engineering and Technology	170	32.76	32.76	41.04
	Social Sciences	204	39.31	39.31	80.35
	Humanities and Arts	100	19.27	19.27	99.61
	Others	2	0.39	0.39	100
School level	Double-First-Class Universities	250	48.2	48.2	48.2
	Non-Double-First-Class Universities	269	51.8	51.8	100.0

Notes Double First-Class Universities Initiative refers to a national plan launched by the Chinese government to support top universities and disciplines. The term “Double First-Class” specifically denotes institutions recognized for either being among the world’s best universities or possessing first-class academic disciplines. A university need only meet one of these criteria to be classified as a “Double First-Class University” in China. Conversely, Non-Double First-Class Universities refer to higher education institutions that have not been included in China’s “Double First-Class Initiative” list

cally. The discussion was audio-recorded with participants' consent and transcribed verbatim for analysis.

Data Analysis

Statistical analyses of the quantitative data, including correlation and regression analyses, were conducted using SPSS software to validate the hypotheses.

Thematic analysis was employed to identify recurring patterns and themes within the qualitative data (Braun & Clarke, 2006). First, open coding was applied to the interview transcripts, followed by axial coding to categorize codes into broader themes (Table 2). These themes were then cross-referenced with quantitative survey results to enhance interpretive validity.

Results

Common Method Variance

The data for this study were self-reported by the participants, which may introduce common method bias (Podsakoff et al., 2024). Therefore, this study used methods such as heterogeneous question forms and reverse scoring to control for possible common method biases (Tepper & Tepper, 1993). The Harman's single-factor test was used to assess common method bias, and the results showed that there were three factors with eigenvalues greater than 1 after unrotated factor analysis. The variance explained by the first common factor was 29.763%, which is below the critical threshold of 40%, indicating that there is no significant issue of common method bias in this study.

Correlation Analysis

A correlation analysis was conducted for each variable (Table 3). The results showed a significant positive correlation between the intention to disclose AI-generated outputs and academic integrity perceptions, and significant negative correlations between the intention to disclose AI-generated outputs and AI usage frequency, as well as between the intention to disclose AI-generated outputs and AI ethics attention. There was a significant negative correlation between academic integrity perceptions and AI usage frequency, and a significant positive correlation between academic integrity perceptions and AI ethics attention. Additionally, there was a significant positive correlation between AI ethics attention and AI usage frequency.

Hypothesis Testing

The SPSS v.14 (Hayes, 2017) was used to process the data and conduct tests on the direct effects, mediating effects, and moderating effects of the regression equations (Table 4).

Regarding the direct effect, the results showed (Table 4) that academic integrity perceptions did not have a significant direct impact on the intention to disclose AI-generated outputs ($\beta=0.041, p=0.351>0.05$). Further, using the percentile Bootstrap method to analyze

Table 2 Coding matrix for qualitative data

Original Citation(Part of screening)	Open Coding	Axial Coding	Broader Theme
"Honestly, when it comes to using AI tools for schoolwork, there's a lot of gray area... There's a lot of confusion about these things, especially since AI is still pretty new in academic settings." (A02)	- Ambiguity in AI use boundaries - Uncertainty due to novelty of AI	Perceived Integrity vs. Disclosure Conflict	Ethical Ambiguity Weakens Direct Effect of Integrity Perceptions
"Even if I feel strongly about academic integrity, I might hesitate to disclose that I used AI... the lack of clear guidelines makes me unsure..." (A05)	- Conflict between integrity and disclosure - Lack of institutional clarity		
"Now I use AI almost every day. It doesn't seem special or out of the ordinary anymore.... It's not something I would normally think needs special attention or mention." (A01)	- Routine dependence on AI - Desensitization to AI assistance	Habitual Use and Ethical Desensitization	Normalization and Desensitization to AI Use
"Honestly, it's really confusing when there are no clear rules or guidelines from the university... Do we need to mention every single time we use an AI tool, even for minor things?" (A03)	- Demand for explicit policies - Over-disclosure concerns	Institutional Accountability Gap	Policy Clarity as a Catalyst for Ethical Alignment

Table 3 Correlation coefficients between variables

	M	SD	1	2	3	4
Intention to disclose AI-generated outputs	0.00	1.00	1			
Academic integrity perceptions	0.00	1.00	0.298**	1		
AI usage frequency	2.17	0.636	-0.180**	-0.020	1	
AI ethics attention	3.02	0.720	-0.154**	0.148**	0.224**	1

Notes * $p < 0.05$, ** $p < 0.01$

the direct effect of academic integrity perceptions, it was found (Table 5) that the 95% Bootstrap confidence interval included zero ($[-0.045, 0.126]$), indicating that academic integrity perceptions did not have a significant direct impact on the intention to disclose AI-generated outputs. Hypothesis 1 was not supported.

Regarding the mediating effect, the results showed (Table 4) that academic integrity perceptions had a significant positive impact on AI ethics attention ($\beta = 0.153, p = 0.000 < 0.01$),

Table 4 Summary of regression models

Variable	AI ethics attention				Intention to disclose AI-generated outputs			
	β	SE	t	p	β	SE	t	p
Constant	3.017	0.031	97.564	0.000**	-0.584	0.546	-1.070	0.285
Academic integrity perceptions	0.153	0.031	4.958	0.000**	0.041	0.044	0.933	0.351
AI usage frequency					0.607	0.259	2.341	0.020*
AI ethics attention					0.358	0.178	2.014	0.045*
AI ethics attention * AI usage frequency					-0.273	0.082	-3.338	0.001**
Sample size	519				519			
R^2	0.045				0.070			
Adjusted R^2	0.042				0.061			
F	F (1,517)=24.577, $p=0.000$				F (4,514)=9.625, $p=0.000$			

Notes * $p < 0.05$, ** $p < 0.01$

Table 5 Direct and mediating effects of academic integrity perceptions on the intention to disclose AI-generated outputs

Notes Boot CI refers to the 95% Bootstrap sampling interval; Bootstrap type: Percentile Bootstrap method

	Level	Effect	Boot SE	Boot CILL	Boot CIUL
Direct effect	—	0.041	0.044	-0.045	0.126
The mediating effect of AI ethics attention	Low level(-1SD)	-0.009	0.013	-0.036	0.015
	average level	-0.036	0.013	-0.064	-
	High level(+1SD)	-0.062	0.019	-0.105	-

and AI ethics attention had a significant positive impact on the intention to disclose AI-generated outputs ($\beta = 0.358$, $p = 0.045 < 0.05$), indicating that AI ethics attention mediated the relationship between academic integrity and the intention to disclose AI-generated outputs. Further, using the percentile Bootstrap method to analyze the mediating effect of AI ethics attention, it was found (Table 4) that at low levels of AI ethics attention, the 95% Bootstrap confidence interval included zero $([-0.036, 0.015])$, meaning there was no mediating effect at this level; at average levels, the 95% Bootstrap confidence interval did not include zero $([-0.064, -0.014])$, indicating a mediating effect with an effect size of -0.037; and at high levels, the 95% Bootstrap confidence interval did not include zero $([-0.105, -0.030])$, indicating a mediating effect with an effect size of -0.065. In summary, the mediating effect of AI ethics attention varied at different levels, further confirming the mediating role of AI ethics attention. This suggests that academic integrity perceptions indirectly promote the intention to disclose AI-generated outputs by enhancing AI ethics attention. In other words, individuals with strong academic integrity perceptions are more likely to have high AI ethics attention, which in turn promotes their intention to disclose AI-generated outputs. Hypothesis 2 is supported.

Regarding the moderating effect, the results showed (Table 4) that the product term of AI ethics attention and AI usage frequency had a significant negative impact on the intention to disclose AI-generated outputs ($\beta = -0.273$, $p = 0.001 < 0.01$), indicating that AI usage frequency moderates the relationship between AI ethics attention and the intention to disclose AI-generated outputs. Further, in conjunction with the mediating effect, it can be concluded

that as AI usage frequency increases, the positive effect of AI ethics attention on the intention to disclose AI-generated outputs gradually weakens and may even become negative. Hypothesis 3 is supported.

To further analyze the specific moderating effect of AI usage frequency, we divided it into low and high groups by adding and subtracting one standard deviation from the mean for simple slope tests. The results (Fig. 2) show that the slopes for both groups of AI usage frequency were negative, and the slope for the high AI usage frequency group was significantly lower than that of the low AI usage frequency group, indicating a negative moderating effect of AI usage frequency. Specifically, although the main effect (i.e., the coefficient of AI ethics attention) in the regression model is positive ($\beta=0.358$), relative to low AI usage frequency, under conditions of high AI usage frequency, the relationship between AI ethics attention and the intention to disclose AI-generated outputs becomes more negatively correlated. This indicates that as AI usage frequency increases, the positive influence of AI ethics attention on the intention to disclose AI-generated outputs weakens and transforms into a more pronounced negative impact. In other words, when AI usage frequency is low, the moderating effect is not very prominent, so the main effect (the positive influence of AI ethics attention on the intention to disclose AI-generated outputs) remains dominant. However, when AI usage frequency is high, the moderating effect becomes more apparent, leading to a significant negative impact of AI ethics attention on the intention to disclose AI-generated outputs. Going even further, using the Johnson-Neyman technique to plot mixed slopes (Fig. 3), the results show that when AI usage frequency is greater than or equal to 1.8, the increase in AI usage frequency weakens the positive influence of AI ethics attention on the intention to disclose AI-generated outputs, and may even transform it into a negative influence.

To test the moderated mediation effect, we calculated the indirect effect of college students' academic integrity on the intention to disclose AI-generated outputs through AI ethics attention at different levels of AI usage frequency (mean $\pm$ one standard deviation) using the Process macro (Table 6). It can be seen that the 95% confidence interval for the high

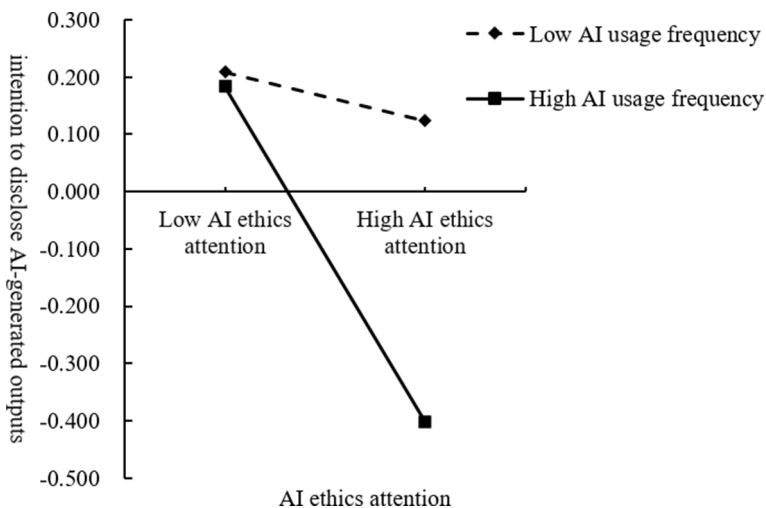


Fig. 2 The moderating effect of AI usage frequency on the relationship between AI ethics attention and the intention to disclose AI-generated outputs (a)

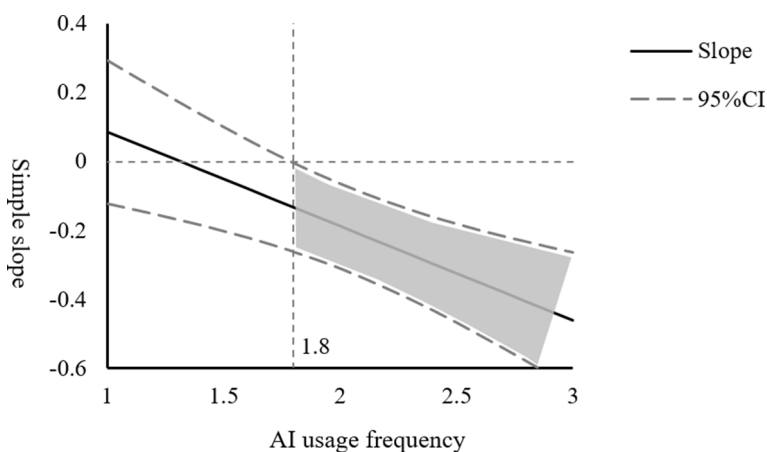


Fig. 3 The moderating effect of AI usage frequency on the relationship between AI ethics attention and the intention to disclose AI-generated outputs (**b**)

Table 6 Results of the moderated mediation effect

Mod- erating variable	Level	Effect	Boot SE	Boot CILL	Boot CIUL
AI usage frequency	High(+1SD)	-0.062	0.019	-0.103	-0.029
	Low(-1SD)	-0.009	0.013	-0.037	0.016
	Group difference	-0.053	0.020	-0.098	-0.017

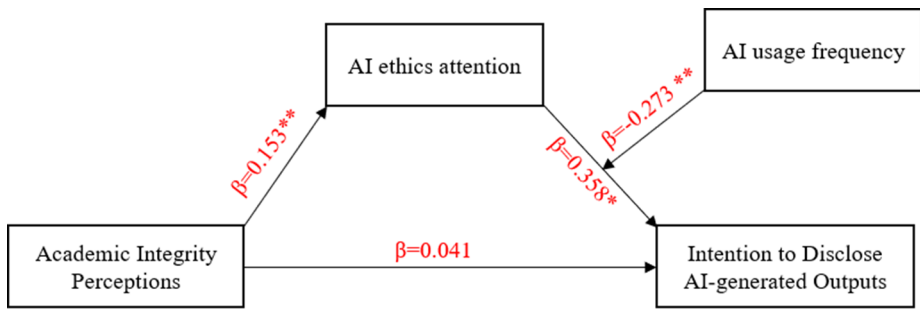
AI usage frequency group does not include zero, while the 95% confidence interval for the low AI usage frequency group includes zero. The difference in effect sizes between the two groups is -0.053, suggesting that the indirect effect of college students' academic integrity on the intention to disclose AI-generated outputs through AI ethics attention is significant in the high AI usage frequency group, but not significant in the low AI usage frequency group. Furthermore, the index of moderated mediation for the model is -0.042, and its 95% confidence interval does not include zero $([-0.077, -0.015])$. This confirms the presence of the moderated mediation effect and further supports Hypothesis 3.

Discussions, Implications and Limitations

Discussions

This study constructs and empirically tests a moderated mediation model (Fig. 4), leading to the following conclusions:

- (1). College students' academic integrity perceptions do not directly predict their intention to disclose AI-generated outputs.
- (2). Academic integrity perceptions indirectly influence college students' intention to disclose AI-generated outputs through the mechanism of AI ethics attention.



Notes: * $p<0.05$, ** $p<0.01$

Fig. 4 Results of the hypothetical test

- (3). The frequency of AI use negatively moderates the relationship between AI ethics attention and the intention to disclose AI-generated outputs; as the frequency of AI use increases, the positive effect of AI ethics attention on disclosure behavior diminishes and transforms into a negative effect.

The Direct Effect of Academic Integrity Perceptions on AI Use Disclosure

Addressing the first research question, *Do college students' perceptions of academic integrity directly predict their intention to disclose AI-generated outputs?*, contrary to our initial hypothesis that academic integrity perceptions would directly boost disclosure intentions, our findings indicate that there is no direct relationship between these two variables. Hypothesis 1 was not supported. This finding contrasts with Social Cognitive Theory's premise that values directly shape behavior (Bandura, 1988). First, strong academic integrity values require an "ethical translator" -AI ethics attention-to convert abstract principles into concrete disclosure behaviors. This extends techno-ethics frameworks (Stahl & Eke, 2024) by identifying a critical cognitive bridge between moral standards and technological practices. Second, the indirect pathway ($\beta=0.153 \rightarrow 0.358$) suggests that in AI-assisted writing contexts, traditional academic integrity frameworks need ethical recalibration. This supports Eaton's (2023) "post-plagiarism" paradigm where conventional integrity metrics fail to capture AI-specific ethical dilemmas.

This finding highlights the complexity of academic integrity in different contexts. Traditional views on academic integrity focus on avoiding plagiarism and data falsification, but disclosing AI-generated content-a newer issue-has not yet been fully integrated into students' perceptions. Consequently, academic integrity education should incorporate AI ethics to enhance understanding of the importance of disclosure.

Moreover, there's a gap between students' academic integrity views and their actual behavior. High integrity perceptions do not guarantee corresponding disclosure actions, possibly due to unclear guidelines or lack of institutional support for disclosing AI-generated outputs.

Our qualitative insights from open-ended responses shed light on this finding. Many students expressed uncertainty about what constitutes "proper" versus "improper" use of AI tools, particularly in academic settings. For instance, one student noted,

“Honestly, when it comes to using AI tools for schoolwork, there’s a lot of gray area. For example, if I use an AI to help me outline an essay, is that okay? Or can I use AI to polish the language and translate my paper? There’s a lot of confusion about these things, especially since AI is still pretty new in academic settings.” (A02).

This ethical ambiguity often led to hesitation in disclosing AI-generated content, even among students with strong academic integrity perceptions. Another student elaborated,

“Even if I feel strongly about academic integrity, I might hesitate to disclose that I used AI. It’s kind of like, ‘Am I doing something wrong?’ or ‘Is this cheating?’ Even if I know deep down that I’m not trying to cheat, the lack of clear guidelines makes me unsure whether I should mention it or not.” (A05).

This suggests that the lack of direct influence may be due to students’ confusion about how to apply traditional academic integrity principles to the emerging context of AI use.

College students’ perceptions of academic integrity and their disclosure intention to AI-generated outputs may reflect developmental differences in moral reasoning schemas. Unlike Kohlberg’s theory of moral development, which strictly defines moral stages by age (Kohlberg, 1975), Rest et al. (2000) proposed a neo-Kohlbergian approach that emphasizes the evolution of cognitive structures and how individuals interpret moral dilemmas through distinct schemas. This means people of various ages can be in any of these schemas. It depends on their personal experiences, educational background, and social environment. The neo-Kohlbergian theory outlines three flexible cognitive schemas: the Personal Interest schema (focusing on direct consequences), the Maintaining Norms schema (adherence to institutional rules), and the Postconventional schema (principled analysis of societal impacts). Students operating within the Personal Interest schema might weigh disclosure decisions based on personal risks (e.g., penalties) or rewards. Therefore, without clear institutional rules or penalties for not disclosing AI-generated outputs, students may not voluntarily disclose such content, even if they have strong perceptions of academic integrity. Those guided by the Maintaining Norms schema would prioritize institutional policies, disclosing AI outputs primarily to fulfill external expectations rather than internalized ethics. Students at this stage might decide whether to disclose AI-generated content based on school policies or teacher requirements. Students in the post-conventional schema have a high level of moral awareness, acting based on internal principles rather than external rules. These students are more likely to choose to disclose AI-generated outputs out of a sense of moral duty because they believe it is the right thing to do.

The Mediating Effect of AI Ethics Attention

Turning to the second research question, How do college students’ perceptions of academic integrity indirectly influence their intention to disclose AI-generated outputs through the mechanism of AI ethics attention?, results show that academic integrity perceptions positively affect AI ethics attention, which in turn influences disclosure intentions. The mediation effect of AI ethics attention helps explain this indirect relationship, supporting Hypothesis 2. This process aligns with but extends Festinger’s (1957) cognitive dissonance theory: while traditional models emphasize post-hoc rationalization of value-behavior gaps, this finding reveals that AI ethics attention acts as a proactive dissonance regulator. Students with higher academic integrity internalize ethical vigilance as a preemptive calibra-

tion mechanism (Macfarlane et al., 2014), filtering AI usage decisions through anticipatory accountability frameworks.

Specifically, college students with higher academic integrity perceptions tend to show greater AI ethics attention. This reflects a schema-driven translation of core academic values into emerging technological contexts—a process resonating with Bandura's () concept of moral agency activation. Unlike passive compliance with institutional norms, such attention embodies active ethical negotiation (Floridi, 2019), where students consciously map integrity principles onto AI's epistemic uncertainties.

The study also shows that AI ethics attention positively impacts the intention to disclose AI-generated outputs. College students with higher AI ethics attention are more likely to disclose AI-generated content in their work. This differs from the findings of Rodrigues et al. (2024): rather than AI tools eroding ethical judgment, heightened ethics attention transforms dissonance into deliberative disclosure practices—effectively countering Eaton's (2023) “post-plagiarism” ambiguity through value-anchored resolution. By framing disclosure as a dissonance reduction strategy, students reconcile their academic identity (“I am honest”) with AI's collaborative role (“I use tools responsibly”)—a dynamic alignment predicted by social cognitive theory.

The Moderating Effect of AI Usage Frequency

Regarding the third research question, How does the frequency of AI use affect the relationship between AI ethics attention and the intention to disclose AI-generated outputs?, we found that increased AI usage frequency negatively moderates this relationship, supporting Hypothesis 3. This suggests that habitual use of AI tools may reduce the perceived necessity for ethical disclosure. This finding reveals a “techno-moral desensitization” process aligning with habituation theory (Waters & McDonald, 1976). Specifically, as the frequency of AI use increases, the positive effect of AI ethics attention on disclosure behavior diminishes and can even become negative. This phenomenon echoes Verbeek's (2011) technological mediation theory: when AI transitions from “tool” to “environment” through habitual use, ethical deliberation becomes embedded in operational inertia rather than conscious choice-making. For instance, one student noted,

“Now I use AI almost every day. It doesn't seem special or out of the ordinary anymore. For example, if I use an AI to help outline my essay or check my grammar, it feels like no big deal. It's not something I would normally think needs special attention or mention.” (A01).

This habituation causes students to underestimate the originality and value of AI outputs, weakening their intention to disclose such content. Additionally, habituation and cognitive adaptation may lead students to view AI-generated content as part of their own knowledge, rather than as external contributions, further reducing disclosure intentions. Frequent AI usage leads to a cultural shift where AI use becomes common and nondisclosure of AI outputs becomes the norm. This change in cultural norms may reduce students' perception of AI ethics' importance, thereby decreasing their likelihood of disclosing AI-generated content. Finally, as AI usage increases, students may experience decision fatigue, which reduces decision-making ability and behavioral control after repeated decisions (Pignatiello et al., 2020). In this scenario, college students tired of deciding whether to disclose AI-generated outputs tend to default to nondisclosure. Frequent use of AI tools can lead to cumulative

decision fatigue, as students constantly balance ethical considerations with efficiency needs. This continuous cognitive effort can deplete students' self-regulatory resources, weakening the positive effect of ethical awareness on promoting transparency and responsibility. In some cases, it may even turn into a negative influence on the intention to disclose AI outputs.

Managerial Implications

Universities: Revisiting and Reforming Academic Integrity Education Systems

Although this study did not find a direct impact of academic integrity on AI output disclosure, it confirmed that academic integrity positively influences AI ethics attention, which then affects disclosure intentions. Therefore, enhancing academic integrity education for college students remains crucial for promoting AI output disclosure. The issue of academic integrity has never been a problem that can only be solved in the classroom, but rather a problem that involves the entire educational institution of universities (Whitley Jr & Keith-Spiegel, 2001). In the post plagiarism era, with the deep integration of generative artificial intelligence and academic creation, traditional academic integrity education is no longer sufficient to meet the challenges brought by emerging technologies. Therefore, universities need to design new strategies for academic integrity education to cultivate students' AI ethics literacy, particularly regarding appropriate disclosure of AI-generated outputs. First, revise the existing academic integrity policy to include "technical clarity" with clear guidelines on new technologies (Perkins & Roe, 2024). Second, revise academic integrity policies to address AI challenges, and update academic norms guidelines to clearly define when AI use must be disclosed and the extent to which AI tools can be used. As one student pointed out,

"Honestly, it's really confusing when there are no clear rules or guidelines from the university about using and disclosing AI-generated work. Like, what exactly should we be telling our professors? Do we need to mention every single time we use an AI tool, even if it's just for minor things like checking grammar?" (A03).

Providing specific examples will help students understand and comply with these rules. Clear guidelines would eliminate much of this confusion and ensure that students know exactly when and how to disclose their use of AI tools, thereby promoting transparency and maintaining academic integrity. Third, organize specialized training sessions or seminars on AI ethics to help students establish healthy AI usage habits. In addition to regular course instruction, universities can hold periodic specialized training sessions or seminars on AI ethics and academic integrity, inviting experts to share insights and using scenario simulations to teach students about AI applications and proper disclosure practices. Fourth, establish a multi-level supervision mechanism that includes self-supervision, peer supervision, and oversight by faculty and management. Self-supervision means that students take responsibility for their own actions; peer supervision encourages mutual reminders and support among peers; while faculty and management oversight is more formal, aiming to ensure academic integrity through institutionalized means. Fifth, use technical means to assist in identifying AI-generated content, utilizing software tools specifically designed to detect AI-generated texts, helping instructors quickly identify which parts of students' submissions

were generated by AI. At the same time, these tools can serve as educational tools, helping students understand the characteristics and limitations of AI-generated content.

Teachers: Regulating College Students' AI Ethics Behavior in Classroom Teaching

Firstly, given the negative moderating effect of AI usage frequency on the relationship between attention to AI ethics and intention to disclose AI-generated outputs, university instructors should regularly assess how students use AI tools in courses. This includes evaluating the frequency of student usage, contexts of use, and purposes, and purposes to ensure that students use these tools appropriately without becoming overly reliant. Secondly, in classroom teaching, teachers should clarify the usage rules and disclosure requirements for AI tools. This includes both school-wide policies and specific course requirements. Specifically, it is necessary for teachers to clearly list these regulations in the course syllabus (Dabis & Csáki, 2024), clearly indicate which AI tool usage constitutes academic misconduct (Cavazos et al., 2024), and promote them to students at the beginning of the course, emphasizing their supervision and disciplinary measures, in order to build an honest classroom (Burden, 2020). Furthermore, to enhance students' awareness of AI ethics, instructors can include activities like group discussions, case analyses, and role-playing exercises in the classroom (Lacković & Olteanu, 2023). These activities help students intuitively understand AI ethics and make informed judgments in real-world scenarios. Lastly, instructors can guide students to document their AI tool usage and reflect on its impact on their learning, thereby strengthening students' self-monitoring capabilities. For example, students could be asked to write brief reflections after each instance of using an AI tool to assist in generating content, explaining the purpose, method of use, and whether appropriate disclosure was made. In summary, to regulate college students' ethical AI behavior, it is necessary to play the role of teachers as "gatekeepers" and effectively supervise the learning process of students (Kim et al., 2022).

Academic Institutions: Establishing Clear Policies for Disclosing AI-Generated Outputs

As previously mentioned, for college students whose moral development is at the pre-conventional level, even those with strong academic integrity may lack motivation to voluntarily disclose AI-generated content if there are no explicit institutional guidelines or penalties requiring acknowledgment of AI contributions. Currently, academic publishing institutions in China have few clear policies on disclosing AI-generated outputs. Therefore, academic institutions should develop a set of clear policies for disclosing AI-generated outputs to guide college students on how to properly disclose when submitting works containing AI-generated content. These policies should detail the circumstances under which AI use must be disclosed and how to do so. Additionally, the policies should emphasize the consequences of non-disclosure or improper disclosure to increase students' awareness of its importance.

Limitations and Future Research

Although this study has yielded valuable findings, it also has certain limitations. Firstly, the sample size could be further expanded to enhance representativeness and more accurately

reflect the characteristics of the overall population of college students, thereby reducing bias and random error and making the research conclusions more reliable. Secondly, the questionnaire used in this study was self-constructed, and while it met statistical standards for reliability and validity, its internal consistency was not particularly high. Further refinement is needed to improve its reliability. Additionally, although the moderated mediation model used in this study was significant, the explained variance in the total variation was relatively low. Thirdly, future studies could employ mixed-methods approaches, combining quantitative surveys with qualitative interviews or focus groups, to explore the nuanced experiences and perceptions of students regarding AI use and disclosure. Lastly, this study did not consider the potential impact of differences in cultural backgrounds on the results. Future research could explore these relationships through cross-cultural comparative studies to gain deeper insights into how perceptions of academic integrity and intentions regarding disclosure of AI-generated outputs vary across different cultural contexts.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10805-025-09622-2>.

Acknowledgements Not Applicable.

Author Contributions All authors contributed equally to the following manuscript.

Funding This study didn't receive any funding's.

Data Availability Data will be made available on request.

Declarations

Competing Interests The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

- Artificial Intelligence Act (2024). European Parliament legislative resolution of 13 March 2024 on the proposal for a regulation of the European Parliament and of the Council on laying down harmonised rules on Artificial Intelligence (Artificial Intelligence Act) and amending certain Union Legislative Acts (COM (2021) 0206–C9-0146/2021–2021/0106 (COD)).
- Bahroun, Z., Anane, C., Ahmed, V., & Zacca, A. (2023). Transforming education: A comprehensive review of generative artificial intelligence in educational settings through bibliometric and content analysis. *Sustainability*, 15(17), 12983.
- Bandura, A. (1988). Organisational applications of social cognitive theory. *Australian Journal of Management*, 13(2), 275–302.
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101.
- Burden, P. R. (2020). *Classroom management: Creating a successful K-12 learning community*. Wiley.
- Cavazos, J. T., Hauck, K. A., Baskin, H. M., & Bain, C. M. (2024). ChatGPT goes to college: Exploring student perspectives on artificial intelligence in the classroom. *Teaching of Psychology*, 00986283241268829.
- Chan, C. K. Y., & Tsi, L. H. (2024). Will generative AI replace teachers in higher education? A study of teacher and student perceptions. *Studies in Educational Evaluation*, 83, 101395.
- Colwill, R. M., Lattal, K. M., Whitlow Jr, J., & Delamater, A. R. (2023). Habituation: It's not what you think it is. *Behavioural Processes*, 207, 104845.
- Dabis, A., & Csáki, C. (2024). AI and ethics: Investigating the first policy responses of higher education institutions to the challenge of generative AI. *Humanities and Social Sciences Communications*, 11(1), 1–13.

- Dai, Y., Liu, A., & Lim, C. P. (2023). Reconceptualizing ChatGPT and generative AI as a student-driven innovation in higher education. *Procedia CIRP*, 119, 84–90.
- Dang, J., Xiao, S., & Dewitte, S. (2014). Self-control depletion is more than motivational switch from work to fun: The indispensable role of cognitive adaptation. *Frontiers in Psychology*, 5, 933.
- Dawson, P. (2020). *Defending assessment security in a digital world: Preventing e-cheating and supporting academic integrity in higher education*. Routledge.
- Duah, J. E., & McGivern, P. (2024). How generative artificial intelligence has blurred notions of authorial identity and academic norms in higher education, necessitating clear university usage policies. *The International Journal of Information and Learning Technology*, 41(2), 180–193.
- Eaton, S. E. (2023). Postplagiarism: Transdisciplinary ethics and integrity in the age of artificial intelligence and neurotechnology. *International Journal for Educational Integrity*, 19(1), 23.
- Eke, D. O. (2023). ChatGPT and the rise of generative AI: Threat to academic integrity? *Journal of Responsible Technology*, 13, 100060.
- El Ali, A., Venkatraj, K. P., Morosoli, S., Naudts, L., Helberger, N., & Cesar, P. (2024). Transparent AI Disclosure Obligations: Who, What, When, Where, Why, How. Extended Abstracts of the CHI Conference on Human Factors in Computing Systems.
- Festinger, L. (1957). A theory of cognitive dissonance. Row, Peterson.
- Ganjavi, C., Eppler, M. B., Pekcan, A., Biedermann, B., Abreu, A., Collins, G. S., Gill, I. S., & Cacciamani, G. E. (2023). Bibliometric Analysis of Publisher and Journal Instructions to Authors on Generative-AI in Academic and Scientific Publishing. *arXiv preprint arXiv:2307.11918*.
- Hayes, A. F. (2017). *Introduction to mediation, moderation, and conditional process analysis: A regression-based approach*. Guilford.
- Hmoud, M., Swait, H., Hamad, N., Karram, O., & Daher, W. (2024). Higher education students' task motivation in The generative artificial intelligence context: The case of Chatgpt. *Information*, 15(1), 33.
- Kasneji, E., Seßler, K., Küchemann, S., Bannert, M., Dementieva, D., Fischer, F., Gasser, U., Groh, G., Günnemann, S., & Hüllermeier, E. (2023). ChatGPT for good? On opportunities and challenges of large Language models for education. *Learning and Individual Differences*, 103, 102274.
- Kim, J., Lee, H., & Cho, Y. H. (2022). Learning design to support student-AI collaboration: Perspectives of leading teachers for AI in education. *Education and Information Technologies*, 27(5), 6069–6104.
- Lacković, N., & Olteanu, A. (2023). *Relational and multimodal higher education: Digital, social and environmental perspectives*. Taylor & Francis.
- Liang, J., Wang, L., Luo, J., Yan, Y., & Fan, C. (2023). The relationship between student interaction with generative artificial intelligence and learning achievement: Serial mediating roles of self-efficacy and cognitive engagement. *Frontiers in Psychology*, 14, 1285392.
- Lo, C. K. (2023). What is the impact of ChatGPT on education? A rapid review of the literature. *Education Sciences*, 13(4), 410.
- Macfarlane, B., Zhang, J., & Pun, A. (2014). Academic integrity: A review of the literature. *Studies in Higher Education*, 39(2), 339–358.
- Mulenga, R., & Shilongo, H. (2024). Academic integrity in higher education: Understanding and addressing plagiarism. *Acta Pedagogica Asiana*, 3(1), 30–43–30–43.
- Oubibi, M., Fute, A., Kangwa, D., Alcardo, A. B., & Adarkwah, M. A. (2024). Interactive technologies in online teacher education in Africa: A systematic review 2014–2024. *Education Sciences*, 14(11), 1188.
- Oubibi, M., Hryshayeva, K., & Huang, R. (2025). Enhancing postgraduate digital academic writing proficiency: The interplay of artificial intelligence tools and ChatGPT. *Interactive Learning Environments*, 1–19.
- Perkins, M., & Roe, J. (2024). Decoding academic integrity policies: A corpus linguistics investigation of AI and other technological threats. *Higher Education Policy*, 37(3), 633–653.
- Perkins, M., Gezgin, U. B., & Roe, J. (2020). Reducing plagiarism through academic misconduct education. *International Journal for Educational Integrity*, 16, 1–15.
- Pignatiello, G. A., Martin, R. J., & Hickman Jr, R. L. (2020). Decision fatigue: A conceptual analysis. *Journal of Health Psychology*, 25(1), 123–135.
- Podsakoff, P. M., Podsakoff, N. P., Williams, L. J., Huang, C., & Yang, J. (2024). Common method bias: It's bad, it's complex, it's widespread, and it's not easy to fix. *Annual Review of Organizational Psychology and Organizational Behavior*, 11(1), 17–61.
- Rodrigues, M., Silva, R., Borges, A. P., Franco, M., & Oliveira, C. (2024). Artificial intelligence: threat or asset to academic integrity? A bibliometric analysis. *Kybernetes*.
- Shahzad, M. F., Xu, S., & Zahid, H. (2024). Exploring the impact of generative AI-based technologies on learning performance through self-efficacy, fairness & ethics, creativity, and trust in higher education. *Education and information technologies*, 1–26.
- Stahl, B. C., & Eke, D. (2024). The ethics of ChatGPT—Exploring the ethical issues of an emerging technology. *International Journal of Information Management*, 74, 102700.

- Tepper, B. J., & Tepper, K. (1993). The effects of method variance within measures. *The Journal of Psychology*, 127(3), 293–302.
- Tlili, A., Shehata, B., Adarkwah, M. A., Bozkurt, A., Hickey, D. T., Huang, R., & Agyemang, B. (2023). What if the devil is my guardian Angel: ChatGPT as a case study of using chatbots in education. *Smart Learning Environments*, 10(1), 15.
- Wang, C., Liu, S., Yang, H., Guo, J., Wu, Y., & Liu, J. (2023). Ethical considerations of using ChatGPT in health care. *Journal of Medical Internet Research*, 25, e48009.
- Waters, W. F., & McDonald, D. G. (1976). Repeated habituation and overhabituation of the orienting response. *Psychophysiology*, 13(3), 231–235.
- Whitley Jr, B. E., & Keith-Spiegel, P. (2001). Academic integrity as an institutional issue. *Ethics & Behavior*, 11(3), 325–342.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.